
Tableau Calculations and Data Analysis
Data Foundation
Data Layer
Sources: Excel, CSV, databases, and cloud services
Tables: Rows and columns structure
Relationships: Connect tables using common fields
Joins: Inner, Left, Right, and Full Outer
Unions: Combine rows from similar tables
Field Types
Dimensions: Categorize qualitative data (Blue)
Measures: Quantify numerical data (Green)
Measure Values: Display multiple measures simultaneously
Measure Names: Identifies measure labels
Data Organization
Groups: Combine dimension members into categories
Sets: Define dynamic subsets displayed as IN or OUT
Sorting: Ascending, descending, alphabetical, and measure-based
Filtering: Dimension, measure, date, wildcard, and context filters
Aggregate Functions
Basic Summary
SUM(): Calculates total of numeric values
AVG(): Calculates arithmetic mean
MIN(): Returns smallest value
MAX(): Returns largest value
Counting and Statistics
COUNT(): Counts non-null values
COUNTD(): Counts distinct non-null values
MEDIAN(): Returns middle value
Standard Deviation: STDEV() and STDEVP()
Variance: VAR() and VARP()
Specialized Functions
ATTR(): Returns value if one distinct non-null value exists
COLLECT(): Combines spatial values into spatial object
Usage Rules
Data Types: Require compatible numeric types
Mixing Error: Do not mix aggregate and non-aggregate expressions
Vs Table Calculations: LOD summary vs view-level cumulative results
Statistical Calculations
Central Tendency
Mean: Sum divided by observation count
Median: Middle value in sorted order
Mode: Most frequently occurring value
Ranking Functions
RANK(): Standard ranking with skipped numbers on ties
RANK_DENSE(): Ranking without gaps
RANK_PERCENTILE(): Relative rank percentile
RANK_UNIQUE(): Unique ranking assignment
Percentile and IQR
PERCENTILE(): Calculates relative position
IQR Formula: Q3 - Q1 spread of middle 50%
Outlier Fences: Lower = Q1 - 1.5*IQR, Upper = Q3 + 1.5*IQR
Box-and-Whisker Plot: Visual representation of distribution
Logical Functions
Conditional Statements
IF-THEN-ELSE: Evaluates condition and returns result
IF-ELSEIF-ELSE: Evaluates multiple conditions sequentially
IF(): Inline conditional evaluation
CASE: Compares expression against multiple values
Boolean Operators
AND: TRUE when both conditions are true
OR: TRUE when at least one condition is true
NOT: Reverses logical result
Null Handling
ISNULL(): Checks if field contains NULL
IFNULL(): Replaces NULL with alternative value
ZN(): Replaces numeric NULL with zero
Date Functions
Current Date and Time
TODAY(): Returns current date
NOW(): Returns current date and time
Component Extraction
YEAR(), MONTH(), DAY(): Extracts date components
DATEPART(): Extracts date part as integer
DATENAME(): Returns date part as text name
Date Math and Manipulation
DATEADD(): Adds or subtracts interval from date
DATEDIFF(): Calculates difference between two dates
DATETRUNC(): Truncates date to beginning of period
MAKEDATE(): Creates date from year, month, day
Number Functions
Rounding and Absolute
ABS(): Returns absolute positive value
ROUND(): Rounds to specified decimals
CEILING(): Smallest integer greater or equal
FLOOR(): Largest integer less or equal
Powers and Roots
SQRT(): Calculates square root
SQUARE(): Calculates square of number
POWER(): Raises number to specified power
Arithmetic Operations
MOD(): Returns division remainder
DIV(): Returns integer quotient
SIGN(): Returns -1, 0, or 1
String Functions
Case Conversion
UPPER(): Converts text to uppercase
LOWER(): Converts text to lowercase
PROPER(): Capitalizes first letter of each word
Trimming and Extraction
TRIM(): Removes leading and trailing spaces
LEN(): Returns character count
LEFT() and RIGHT(): Extracts characters from sides
MID(): Extracts substring from position
SPLIT(): Extracts section based on delimiter
Search and Modification
REPLACE(): Replaces substring
CONTAINS(): Checks for substring presence
FIND(): Returns substring position starting at 1